Limits of Confidence-Threshold Repair in External Lung-Opacity Detection: A Multi-Run RSNA–VinDr Study

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Abstract

Background: Severe external detection failure may invite confidence-cutoff adjustment, but a cutoff cannot recover every missed lesion. We examined the limits of this repair under fixed false-positive budgets and retained outputs from two deep-learning lung-opacity pipelines.

Methods: This retrospective study used 5,000 RSNA radiographs in patient-disjoint training, validation, and internal-testing partitions (3,500/750/750). Faster R-CNN and YOLO11s shared canonical inputs, disabled stochastic augmentation, and a common evaluator. All five training runs per detector were retained. A protocol frozen before external inference applied the same checkpoints without adaptation to 3,000 VinDr-CXR images, including 84 images with 95 strict-target boxes. Historical three-run RSNA validation thresholds remained primary; five-run selection was post-hoc sensitivity. Primary uncertainty resampled held-out observations and independent detector runs. Timing used a matched boundary on an RTX 4060 laptop. Subsequent post-hoc analyses decomposed retained-candidate errors, inspected overlap with other local findings, reweighted internal positive frequency, and estimated changes with the same checkpoints coupled across datasets. A separate exploratory diagnostic tested fixed retention and display controls. A subsequent repairability audit jointly estimated cutoff and geometric ceilings on both complete cohorts and verified their interpretation with controlled output changes.

Results: Mean AP@0.50 fell from 0.304224 to 0.002505 for Faster R-CNN and from 0.162612 to 0.000501 for YOLO11s. External AP contrasts retained positive marginal intervals, but historical-threshold recall fell from 0.350746/0.194776 to 0.018947/0.004211, respectively. External precision was 0.010830/0.004271; all historical-threshold metric contrast intervals included zero. Observed FROC means retained the internal ordering, but most external contrast intervals included zero. Missing candidate support allowed an ordering reversal beyond the observed frontier at the upper budgets; detection-cap saturation imposed an additional limit. One low-score YOLO run emitted nothing at the historical threshold and remained included. YOLO11s had 2.75-fold higher measured throughput under the matched timing protocol. Post-hoc diagnostics showed different mixtures of below-threshold localization and missing retained candidates. Other-finding overlap and population weighting informed sensitivity without identifying causes of external failure. At 0.25 FP/image, external cutoff ceilings were 0.086316/0.033684; unavailable retained geometry accounted for 0.431579/0.818947 of reference boxes, respectively.

Conclusions: Cutoff adjustment alone could not restore external recall within the stated budget and retained support. Preserved observed AP ordering did not imply preserved performance or successful operating-point transport. Ranking, score scale, training variability, false-positive regimes, and implementation efficiency require separate interpretation. This is cross-dataset and cross-annotation- ontology transportability testing, with remaining confounding among dataset and pipeline factors; it does not establish diagnostic utility. Code and aggregate results are public; source-data access is governed by provider terms, and trained checkpoints are not publicly released.

1. Introduction

Detector comparisons often summarize performance with average precision (AP) and precision or recall at a chosen confidence cutoff. These answer different questions: AP summarizes a ranked detection set, whereas a cutoff determines which detections are emitted at an operating point. Equal numerical cutoffs need not produce equal selectivity across detectors. Prior calibration work already identifies this problem and the dependence of detector assessment on threshold and metric design (Kuzucu et al. 2024).

We examine its practical consequences for localization of the RSNA challenge category `Lung Opacity`. The annotations describe possible pneumonia-like pulmonary opacities rather than clinically confirmed pneumonia (Shih et al. 2019). Our intended use is methodological comparison of research pipelines, not diagnosis or patient prioritization.

The central research question is: after external failure, how much recall could a global confidence cutoff recover within a fixed false-positive budget, and how much remains constrained by candidate ranking or unavailable retained geometry? This follow-up question was formulated after the original external results. The original comparison examined whether conclusions from ranking, shared raw thresholds and validation-selected operating points transported under a non-identical opacity ontology. We compare Faster R-CNN and YOLO11s using common data, canonical preprocessing, no stochastic augmentation, a common evaluator, and five retained training runs per detector. Patient-cluster and detector-run resampling characterize uncertainty; standardized timing describes the measured implementation costs.

The contribution is a controlled separation of these evaluation objects under a leakage-aware internal protocol and frozen external testing. The RSNA-trained checkpoints and operating rules are applied to VinDr-CXR without adaptation; the question concerns both relative ordering and absolute performance. We do not claim the first discovery of detector score-scale mismatch. Optimization and numerical-precision differences remain disclosed, so conclusions apply to the evaluated pipelines. Secondary calibration, stress-test, and explanation analyses are indexed in the supplement. The internal analysis is retrospective and was not preregistered. The external protocol was locally frozen before full inference, not publicly preregistered.

The subsequent exploratory contribution is a reproducible audit of threshold repairability: exact cutoff envelopes, maximum compatible geometric coverage, joint observation/run uncertainty across domains, and controlled output perturbations that distinguish score-order changes from lost geometry. Earlier annotation-overlap, population-weight and retention sensitivities support its interpretation. The contribution is empirical and diagnostic; it preserves the frozen target and separates measurable repair limits from unidentified causes.

2. Related Work

2.1 Detector scores and operating-point assessment

Küppers et al. introduced multivariate detection-confidence calibration, including dependence on predicted location and box scale (Küppers et al. 2020). Kuzucu et al. subsequently showed how shared score thresholds and inconsistent detection populations can distort joint accuracy and calibration comparisons; AP alone does not select an operating threshold (Kuzucu et al. 2024). This motivates separate reporting of AP/PR, validation-selected thresholds, and FROC here. We evaluate existing prediction sets and fit no calibration model. Supplementary D-ECE is a descriptive, detection-level statistic of the emitted-detection population, dependent on support and binning; it is not clinical-risk calibration.

2.2 RSNA methodology and direct detector comparisons

The RSNA resource augments NIH chest radiographs with expert opacity boxes and categorical labels, including an adjudication process (Shih et al. 2019). Its official dataset description identifies the radiologist annotation workflow and conversion to DICOM with limited demographic and projection tags (Radiological Society of North America 2022). The challenge evaluated a custom image-aggregated score across IoU thresholds 0.40–0.75 (Radiological Society of North America 2018). Our internal COCO AP protocol uses different matching/aggregation and IoU ranges; its values are not challenge leaderboard scores.

Direct RSNA comparisons already exist. Wu et al. compare an anchor-free detector with SSD, Faster R-CNN, RetinaNet, and FCOS, alongside augmentation and focal-loss ablations (Wu et al. 2024). More recently, Chinnam et al. compare YOLOv8l, Faster R-CNN with FPN, and Deformable DETR on RSNA, reporting accuracy and speed and selecting a confidence cutoff using validation (Chinnam et al. 2026). Their reported input resolutions differ by detector; their split, model scales, and evaluation choices also differ from ours.

Neither paper's numerical results are directly interchangeable with this benchmark. The present focus is the effect of operating-point definition and training variability within two disclosed pipelines, rather than a new claim that detector comparisons are absent.

2.3 Implementation and reporting context

Faster R-CNN combines region proposals with a second-stage classifier and box regressor (Ren et al. 2015); FPN supplies multiscale features (Lin et al. 2017). The YOLO11s implementation is identified by its pinned Ultralytics release and model configuration (Ultralytics 2026a, 2026b). These specify the compared implementations without implying a causal effect of detector family.

CLAIM 2024 recommends explicit data partitions, reference standards, uncertainty, reproducibility, and reporting of unavailable information (Tejani et al. 2024). We use its distinction between internal and external testing; `validation` below denotes the model-optimization partition used for checkpoint and threshold selection. The accompanying reporting crosswalk records remaining gaps and is not a certification of compliance.

2.4 External testing under a different annotation ontology

VinDr-CXR provides radiologist-annotated local findings and separate global labels (Nguyen et al. 2022). Its local lung-opacity category is not an identical reference standard to RSNA's pneumonia-like opacity target. External testing therefore assesses cross-dataset and cross-annotation-ontology transportability. Holding checkpoints and operating rules fixed allows ranking transport to be examined separately from the transport of absolute performance and emissions. It does not isolate which dataset or implementation difference causes a shift.

Zech et al. demonstrated variable cross-hospital chest-radiograph performance and dataset-associated signals in pneumonia classification (Zech et al. 2018); that evidence motivates external testing without providing a localization benchmark interchangeable with ours. Reinke et al. describe how prevalence, reference definitions and aggregation affect image analysis metrics (Reinke et al. 2024). Positive-frequency standardization here is a sensitivity calculation, not a fitted label-shift correction: label-shift methods rely on assumptions about stable conditional distributions that are not established across these datasets (Lipton et al. 2018). Training-run resampling reflects benchmark variability (Bouthillier et al. 2021), while FROC describes lesion localization against false-positive burden (Chakraborty 2010). None of these established ideas is claimed as a new method; the contribution is their joint, reproducible empirical use.

2.5 Error diagnostics and the incremental contribution

TIDE separates detection error types and measures their effects on AP through diagnostic corrections (Bolya et al. 2020). LRP combines localisation, false-positive and false-negative errors and defines an optimal score threshold (Oksuz et al. 2021). Calibration research additionally distinguishes probability quality from detection accuracy (Kuzucu et al. 2024). These establish that error diagnosis and threshold analysis are existing research areas.

Exact FROC already provides the best retained recall at each FP/image budget. Our audit places that endpoint alongside maximum geometric matching and joint cross-domain uncertainty, then tests the interpretation using controlled changes to saved outputs. It answers a restricted repair question without introducing a new accuracy metric, a new invariance theorem, or an empirical claim of superiority over TIDE or LRP. No such benchmark comparison was performed.

3. Materials and Methods

3.1 Dataset, target, and patient-disjoint split

We used the Stage 2 training set from the 2018 RSNA Pneumonia Detection Challenge. The complete source contains 26,684 labeled radiographs and 9,555 positive boxes. The single foreground category was `Lung Opacity`; `Normal` and `No Lung Opacity / Not Normal` remained zero-box negative studies.

Compute constraints motivated a deterministic 5,000-study subset, without a formal sample-size or power calculation. Selection used seed 17, SHA-256 ordering, and the three label strata while keeping each NIH patient group together and tracking source stratum proportions. The other 21,684 studies were excluded for compute scope. The Kaggle `patientId` identifies an examination, so the official RSNA mapping supplied

NIH patient keys. Selected studies came from 2,136 patient groups; train/validation/internal-test patient-key intersections were empty.

For cohort description, split manifests were matched to original DICOM files. Header-only extraction retained aggregate age, sex, and projection counts. All 5,000 age elements had VR `AS` but bare numeric values without required D/W/M/Y units. Values from 0 to 120 were interpreted as nominal years; one out-of-range value was excluded. Quartiles used linear percentiles. Sex and projection were counted as encoded, with percentages based on all studies in each partition.

Table 1. RSNA cohort and limited DICOM-header characteristics by partition.

Characteristic	Training	Validation	Internal testing	Total
Studies	3,500	750	750	5,000
Patient groups	1,492	321	323	2,136
Opacity-positive, n (%)	798 (22.8%)	169 (22.5%)	169 (22.5%)	1,136 (22.7%)
Reference boxes	1,267	277	268	1,812
Age, median [IQR], nominal y	50 [36–60]	49 [34–59]	47 [35–61]	49 [36–60]
Age summary, n	3,499	750	750	4,999
Age unavailable, n (%)	1 (0.03%)	0 (0.0%)	0 (0.0%)	1 (0.02%)
Female, n (%)	1,614 (46.1%)	399 (53.2%)	349 (46.5%)	2,362 (47.2%)
Male, n (%)	1,886 (53.9%)	351 (46.8%)	401 (53.5%)	2,638 (52.8%)
AP, n (%)	1,690 (48.3%)	348 (46.4%)	325 (43.3%)	2,363 (47.3%)
PA, n (%)	1,810 (51.7%)	402 (53.6%)	425 (56.7%)	2,637 (52.7%)
Sex / projection missing, n	0 / 0	0 / 0	0 / 0	0 / 0

No other sex or projection category was observed. These header fields are study-level descriptors, not independently verified clinical demographics.

Canonical preprocessing

The annotation audit found no malformed, non-positive-area, off-image, or exact duplicate positive boxes. One conversion path checked DICOM pixel arrays for finite content, inverted `MONOCHROME1` where needed, and applied per-image min-max scaling to 8-bit grayscale PNG. COCO annotations were canonical; the YOLO labels were derived from the same records. Both pipelines used 640-pixel inputs and their disclosed native resizing paths. Source hashes and split records are indexed in Supplementary Section S1.

3.2 Detector pipelines and controlled training factors

The models were Torchvision `fasterrcnn_resnet50_fpn_v2` and Ultralytics YOLO11s (`8.4.110`), both initialized from COCO weights and adapted to the single foreground class. Both used SGD, at most 30 epochs, validation mAP@0.5:0.95 checkpoint selection and early stopping after at least eight epochs, and seeds 17, 42, 137, 271, and 314. Stochastic augmentation was disabled, including the Ultralytics-only mosaic, mixup, cutmix, copy-paste, color, flip, geometric, erasing, and multi-scale transforms.

Faster R-CNN used physical batch 2 with two-step accumulation, float16 AMP, frozen pretrained BatchNorm statistics, learning rate 0.005, and a validation-plateau scheduler. YOLO11s used batch 4, bfloat16 forward/backward with float32 assignment/loss, native BatchNorm updates, and one-epoch warmup to a constant learning rate of 0.001. These exceptions followed numerical stability diagnostics before accepted benchmark training. Thus the training budget framework and data controls were common, but optimization, precision, losses, and assignment were not identical. Full configurations are indexed in the implementation record.

3.3 Five retained runs and common evaluation

All five accepted training attempts per detector were retained, including a YOLO11s run with no detections at the shared cutoff. None was replaced because of internal-test behavior. Seed labels identify reproducible runs; they are not matched stochastic blocks across frameworks. Checkpoints were selected using validation mAP before internal testing.

Both adapters emitted source-image `xyxy` boxes, canonical categories, and scores to one evaluator. COCO AP used the official pycocotools protocol (Lin et al. 2014) at IoU 0.50 and averaged over 0.50:0.95, with a retained score floor of 0.001 and at most 100 detections per image. Single-point precision, recall, and F1 were global micro metrics within each run at match IoU 0.50; run summaries used equal-run means and sample SDs. The matcher processed predictions in stable descending-score order, choosing the highest-IoU unmatched same-class reference without reuse. Native NMS used IoU 0.50. Zero detections contributed zero precision, recall, and F1 by the declared convention. Mean matched-box IoU and Dice were undefined without a true positive and excluded misses; they used five Faster R-CNN and four YOLO11s runs and remain secondary localization diagnostics.

3.4 Operating-point protocols and exact-score FROC

The shared cutoff of 0.25 was a protocol-sensitivity endpoint, not a deployment choice. Five-run exploratory test sweeps used 99 thresholds from 0.01 to 0.99. Official PR curves came from the COCO interpolated precision tensor at 101 recall positions, independently of this threshold grid. Test curve optima were descriptive and did not select operating thresholds.

Historical detector-specific thresholds maximized arithmetic mean validation F1 across seeds 17, 42, and 137 on the same 99-point grid; exact ties favored the higher threshold. They were frozen and applied unchanged to all five test runs. A separate post-hoc sensitivity applied the identical rule to validation predictions from all five frozen checkpoints per detector, then applied its thresholds unchanged to the same five test runs. Test labels were not used in either selection. The five-run sensitivity was not prospectively frozen and did not replace the historical three-run rule.

FROC evaluated every unique retained score in descending order, plus an empty upper sentinel and the lower candidate endpoint. Separate inference-only prediction collection used the same ten frozen checkpoints, 750 images, annotations, preprocessing, matcher, NMS, and detection cap, with a candidate floor of 0.00001. No retraining or threshold optimization on test outcomes was performed. At each prespecified budget (0.125, 0.25, 0.5, 1, and 2 FP/image), each run contributed its highest observed sensitivity without exceeding that budget. Tied scores entered together; a tied-score group was not partially admitted to fill a budget. No interpolation or extrapolation was used. Means and sample SDs summarized five runs per detector. The frontier remains conditional on the candidate floor; an endpoint below a budget is reported as floor-limited.

3.5 Frozen VinDr external testing

We included the complete official VinDr-CXR v1.0.0 test release obtained through credentialed PhysioNet access (Nguyen et al. 2021; Pollard et al. 2026). The source describes consensus test annotations from three initial radiologists and two reviewers (Nguyen et al. 2022). The strict target concept was `Lung opacity`, mapped only to the exact released local label `Lung Opacity`. Consolidation, Infiltration, Atelectasis, ILD, global Pneumonia, and other findings were not merged. Images without strict-target boxes remained negatives for this task, including images with other abnormalities; those findings were not ignore regions.

The protocol fixed cohort inclusion, ontology, preprocessing, checkpoints, endpoints, candidate floor, cap, and both transported threshold policies before full external inference. For this original frozen evaluation, no VinDr annotations were used for training, fine-tuning, model or checkpoint selection, threshold selection, calibration fitting, or outcome-guided ontology, preprocessing, NMS, or endpoint changes. No VinDr-optimized threshold was created. All ten RSNA-trained checkpoints were applied without adaptation. AP used the common 0.001 evaluation floor; exact-score FROC used the frozen 0.00001 collection floor, IoU-0.50 matching/NMS, and cap of 100 detections per image. Native candidate filters use strict `score > floor`; the common evaluator uses `>=` on the retained candidates. FROC score coordinates were not exported as operating rules.

Source checksums, complete decoding, PNG round trips, source-to-COCO equality, common-loader compatibility, and inverse-coordinate checks preceded inference. External conversion used the same min-max/inversion policy and frozen native resizing, with no test-time augmentation. External inference verified float16 AMP for Faster R-CNN and bfloat16 AMP for YOLO11s. Historical internal YOLO accuracy inference used FP32. Thus internal/external score and performance changes cannot be attributed solely to dataset differences.

A secondary sensitivity subsequently applied the external YOLO numerical path to the same internal images and all retained checkpoints, preserving historical FP32 results. It used explicit bfloat16 autocast and the external disabled-TF32 policy, unchanged native preprocessing/postprocessing, candidate support, evaluator and frozen thresholds. Historical effective TF32 flags were not recorded, so this assesses the numerical path rather than activation dtype alone. Candidate agreement used same-image, same-class greedy descending-IoU

matching; unmatched candidates were retained in the emission comparison. Full settings, paired differences and matching details are in Supplementary Section S10.

A narrow Faster R-CNN numerical correction canonicalized a documented one-ULP float32 inverse-coordinate overshoot at the upper image boundary. Negative lower bounds and materially out-of-range coordinates still fail. The correction preceded any completed external result and changed no frozen scientific setting; it was not post-hoc external tuning. Raw-coordinate replay and the original failed attempt are documented in Supplementary Section S9.

No defensible VinDr patient linkage was available in the released annotations or inspected headers. The external resampling unit was therefore the released image, not a verified independent patient. Repeated patients could make these intervals too narrow.

3.6 Estimands and uncertainty

Primary pointwise 95% bootstrap intervals combined patient sampling and stochastic training variability (Efron and Tibshirani 1993). In 2,000 two-stage draws, NIH patient groups were sampled with replacement, all examinations of each selected patient moved together, and runs were resampled independently within detector. AP and other nonlinear metrics were reconstructed from each resampled prediction set. Precision, recall, F1, and AP used five runs per detector; conditional IoU/Dice used five/four defined runs. The effect estimate was the unstandardized Faster-R-CNN-minus-YOLO11s difference.

Secondary two-sided permutation tests held checkpoints fixed and used 5,000 patient-group detector-label swaps, a plus-one correction (Phipson and Smyth 2010), and Holm adjustment across seven clean endpoints (Holm 1979). Conditional localization used four complete same-label pairs for this secondary test only. These p-values do not test the same training-procedure estimand as the bootstrap intervals. No seed-aware p-value was introduced. Those historical intervals do not incorporate threshold-selection uncertainty and do not cover validation-selected or FROC endpoints.

The external analysis used the same inferential principles, with 2,000 draws separately per dataset. Within each draw, observations were shared across both detectors and all runs; trained runs were sampled independently within detector. RSNA resampled its 323 known patient groups, while VinDr resampled 3,000 released images. Cohort-level AP, exact-score FROC, and historical-threshold precision, recall, F1, and FP/image were reconstructed before averaging runs. The primary estimand concerns the disclosed training procedure conditional on its fixed training data and recipe; training patients were not resampled or retrained. Secondary seed-17 checkpoint-conditional intervals resampled only observations. Same-number detector seeds were not paired stochastic replicates.

Aligned internal intervals for these endpoints were computed separately and preserve the historical analysis above. Percentile intervals are marginal 95% intervals, not simultaneous or familywise guarantees. They condition on the selected thresholds and retained floor/cap support; neither threshold-selection uncertainty nor missing candidate support is resolved. Undefined draws are reported without retrying or imputing. The post-hoc five-run threshold policy and score/emission summaries remain descriptive. Cohorts and predictions were never pooled, and no cross-dataset interaction test was performed in that original analysis. The later post-hoc change intervals are specified in Section 3.9. Labels such as unchanged, strengthened, weakened, and reversed describe observed ordering or raw gaps, not significance decisions. Full estimands and all intervals are in Supplementary Section S9.

3.7 Standardized implementation timing

The primary timer starts from the same decoded uint8 RGB source image in host memory. It includes resize/letterbox, tensor conversion, host-to-device transfer, model forward, native postprocessing/NMS, source-coordinate restoration, and CPU extraction of boxes, scores, and labels. It excludes disk I/O and decoding. CUDA synchronization brackets every timed interval. The protocol uses the same 100 test images ranked by SHA256 of seed 17 and filename, batch 1, and the frozen seed-17 checkpoint per detector. Three full repetitions alternate detector order, each preceded by 10 warm-up images per detector. Model setup, fusion, reference inference, and result checking are outside the timer.

The reporting machine was an ASUS ROG Strix G16 with i7-13650HX, 16 GB RAM, and RTX 4060 Laptop GPU (8 GB). The environment was Windows, Python 3.11.15, Torch 2.6.0+cu124, Torchvision 0.21.0+cu124, Ultralytics 8.4.110, CUDA 12.4, cuDNN 90100, and driver 610.47. Explicit inference AMP used verified float16 convolutions for Faster R-CNN and bfloat16 for YOLO11s. Both models were resident; one CPU

thread, disabled TF32/cuDNN benchmarking, and deterministic algorithms were used. The common floor was 0.001, NMS IoU 0.50, and cap 100.

Timed outputs were checked against ordinary file inference using the same explicit AMP and postprocessing. Agreement is not claimed against the historical FP32 YOLO evaluation bundles, which remain unchanged. FPS divides timed calls by total elapsed time; median and IQR describe image latency. Timing repetitions are technical repeats, not independent biological or training replicates. Full timing conditions and raw checks are in the compute record. Historical asymmetric timing, Pareto panels, training profiles, and incomplete profiler-registered operation counts remain supplementary and are not pooled with this timing protocol.

3.8 Secondary analyses

Detection-level D-ECE (Küppers et al. 2020) used post-NMS predictions at the 0.001 floor and the same IoU-0.50 matcher. Within the one-class stratum, confidence, relative center coordinates, width, and height each used five equal-width bins. Cells with fewer than eight detections contributed zero; all emitted detections remained in the weighting denominator. Thus D-ECE is descriptive and support/binning dependent, conditional on the emitted-detection population, and excludes missed targets. The existing bin/support/floor sensitivity and confidence-only marginal reliability plots are in Supplementary Section S4. No calibration map was fitted; this is not clinical-risk calibration.

Common corruptions, stored-array acquisition/display stress, and Grad-CAM use one checkpoint per detector and a fixed 300-image sample; explanation controls use a nested 50-image sample. Their methods and results are in Supplementary Sections S5–S6. They describe synthetic sensitivity and score-associated maps, not external transportability or causal explanations. Validation-only F-beta and hypothetical linear-loss analyses remain in Supplementary Section S3; their thresholds were not applied to test data. Conventional decision-curve analysis and clinical net benefit were not evaluated.

3.9 Post-hoc external failure and change analyses

After observing external outcomes, a versioned exploratory protocol reused all frozen predictions and historical thresholds. Strict reference boxes were classified in order as matched, unmatched despite an above-threshold compatible candidate, compatible only below threshold, partially localized, or without a relevant retained candidate. Compatibility required IoU at least 0.50; partial localization required best retained IoU from 0.25 to below 0.50. Candidate availability ignores assignment competition and cannot establish recall. Strict false positives were classified in order by overlap with an already claimed strict target, another local finding at IoU at least 0.50, partial local overlap, or no substantial local overlap. Nonexclusive class flags additionally describe Consolidation, Infiltration, Atelectasis and ILD. Global labels were excluded. No broader merged target was scored without expert instance review; overlap never changes a strict false positive into a true positive.

For prevalence sensitivity, internal positive and negative images received weights q/p and (1-q)/(1-p), respectively, where p is internal positive-image frequency and q is the observed external frequency. Every event and reference box inherits its image weight. Weighted precision uses weighted counts; custom weighted AP integrates the interpolated precision envelope on the canonical recall grid. Unit-weight observed AP agrees with canonical COCO AP, but fractional event masses and bootstrap multiplicities define a custom metric, not official COCO AP or replicated-image COCO tie handling. Recall is invariant to this weighting. The calculation preserves internal conditional case mix and cannot recreate external acquisition or annotation changes.

We used 2,000 joint bootstrap draws for external-minus-internal AP, FROC at 0.25 FP/image, historical-threshold recall, and differences in those changes. Patient clusters were sampled internally and released images externally; cohort draws were independent, and each draw was shared across all models within a cohort. Run multiplicities were independent between detector families but identical across domains for each trained checkpoint. The same draws produced prevalence-sensitivity intervals, re-estimating p while holding q fixed. Undefined draws were recorded without replacement. Intervals are marginal and conditional on fixed thresholds, training data and retained support. Full definitions, tests and source hashes are in Supplementary Section S11.

3.10 Subsequent controlled implementation diagnostic

A separate exploratory protocol, designed after the preceding results were known, compared fixed DICOM-header windowing and a factorial change in the collection floor and detection cap. It used every strict-positive

external image and an equally sized, fixed random sample of strict negatives, with all retained checkpoints. Historical and nested baseline predictions had to replay exactly. Maximum one-to-one geometric coverage used bipartite matching and was an annotation-informed upper bound, distinct from canonical threshold recall. Paired bootstrap draws shared images and checkpoints across arms, resampling positive and negative images separately. Marginal intervals were conditional on this diagnostic cohort; negative-control FP/image did not estimate full-cohort FP/image. A separate full-cohort control changed only the AP evaluation floor on the original saved predictions. The protocol, complete arm results and verification are in Supplementary Section S12.

3.11 Fixed-output threshold repairability

The exploratory repairability protocol was specified before its new computations, with the earlier external results already known. It reused every saved run on both complete test cohorts. Original candidate membership, strict target mapping, matching IoU and cap were retained. Baseline FROC, historical counts and external maximum coverage had to replay exactly; the internal lower-floor bundles also reproduced the original historical counts.

For a fixed run, let R_b be maximum greedy-matched recall over exact score cutoffs satisfying FP/image at most b, R_{all} recall with all retained candidates admitted, and G the maximum-cardinality, same-class bipartite matching count divided by the reference-box count at the same IoU. Then

$$1 = R_b + (R_{all} - R_b) + (G - R_{all}) + (1 - G).$$

The terms are the cutoff ceiling, current-order FP-budget barrier, assignment gap and unavailable retained geometry. They account for a fixed-output recall shortfall, not causal shares of cross-domain loss. A strictly increasing global score map preserves prefixes and ties, so cannot raise R_b when membership, matching and postprocessing stay fixed. Feature-dependent reranking, changed ties, NMS, boxes, ontology, floor or cap fall outside this statement. No calibration model was fitted and no maximizing external cutoff was exported. Per-run ceilings are optimistic test-set envelopes; their mean is not the performance of one common cutoff. G is annotation-informed, not a deployed operating point. Historical recall and the signed difference from R_b are reported with budget feasibility; infeasible policies are never discarded.

Joint uncertainty used 2,000 draws. Each draw sampled internal patient clusters and external images independently, shared observation weights across runs and detectors within each cohort, and reused checkpoint multiplicities across domains while sampling the two detector families independently. Nonlinear budget endpoints were recomputed before run averaging. Component and contrast intervals are marginal; finite and undefined draw counts are retained. These new joint intervals do not replace earlier bootstrap analyses.

Controlled copies of every run received global score powers, partial score permutations, or removal of candidates individually compatible or incompatible with any reference. Powers had to preserve entire FP/recall curves; permutations preserved the score multiset and geometric ceiling. Complete compatible removal had to eliminate coverage, while incompatible removal had to preserve geometry and never worsen the frontier. Imposed replicates were averaged within runs; they were not extra patient or training samples. Candidates were not refiltered at the old numerical floor after score transformation. All original budgets, intervention strengths and replicates are retained in Supplementary Section S13.

4. Results

All detector summaries weight the five retained runs equally. AP is on the 0–1 scale, not a percentage. A denotes Faster R-CNN and B denotes YOLO11s. No checkpoint predictions are pooled into an ensemble.

4.1 Clean internal-testing performance

Across the selected cohort, median nominal age was 49 years (IQR 36–60; n=4,999), 2,362 studies (47.2%) were encoded female and 2,638 (52.8%) male, and 2,363 (47.3%) were AP versus 2,637 (52.7%) PA. Sex and projection had no missing or other values. Age had no missing tags, but all age values lacked encoded units and the single out-of-range value was excluded. These are descriptive cohort characteristics; no demographic subgroup comparison or fairness inference was performed.

The unified evaluator processed 750 internal-testing images and 268 reference boxes for each of 10 frozen checkpoints. Both AP endpoints favored Faster R-CNN. At the shared score threshold of 0.25, Faster R-CNN

had higher recall and F1, whereas YOLO11s had higher precision and slightly higher conditional localization among successfully matched detections. YOLO11s conditional IoU and Dice used only four defined seeds.

Table 2. Internal-test endpoints at the shared cutoff and on ranked predictions.

Endpoint	Faster R-CNN, mean +/- SD	YOLO11s, mean +/- SD
Precision at 0.25	0.1959 +/- 0.0552 (n=5)	**0.2983 +/- 0.1691 (n=5)**
Recall at 0.25	**0.5799 +/- 0.0911 (n=5)**	0.0955 +/- 0.0607 (n=5)
F1 at 0.25	**0.2845 +/- 0.0528 (n=5)**	0.1427 +/- 0.0868 (n=5)
Conditional matched-box IoU	0.6749 +/- 0.0065 (n=5)	**0.6985 +/- 0.0157 (n=4)**
Conditional matched-box Dice	0.8010 +/- 0.0049 (n=5)	**0.8181 +/- 0.0111 (n=4)**
mAP@0.5	**0.3042 +/- 0.0189 (n=5)**	0.1626 +/- 0.0162 (n=5)
mAP@0.5:0.95	**0.0995 +/- 0.0067 (n=5)**	0.0542 +/- 0.0060 (n=5)

YOLO11s seed 271 illustrates the distinction between ranking and score scale. Its training losses decreased and validation mAP converged normally, while internal-testing AP@0.5 and AP@0.5:0.95 were 0.15872 and 0.05558, within the other YOLO seed range. Yet its maximum internal-testing confidence was 0.0412735. It therefore emitted no detection at 0.25 and contributed observed zeros to precision, recall, and F1; matched-box IoU and Dice were mathematically undefined. Its low scores changed fixed-threshold behavior without a corresponding AP collapse. The run remained in every analysis where its endpoint was defined.

4.2 Shared thresholds and validation-selected operating points

On the five-run official AP@0.5 curve, Faster R-CNN had higher mean interpolated precision at 97 of 101 recall positions, with four ties. COCO AP ranking does not depend on the selected single operating threshold, but remains conditional on the retained prediction floor and common evaluation/post-processing protocol. The higher YOLO11s precision at 0.25 therefore did not imply a precision advantage at matched recall (Figure 1).

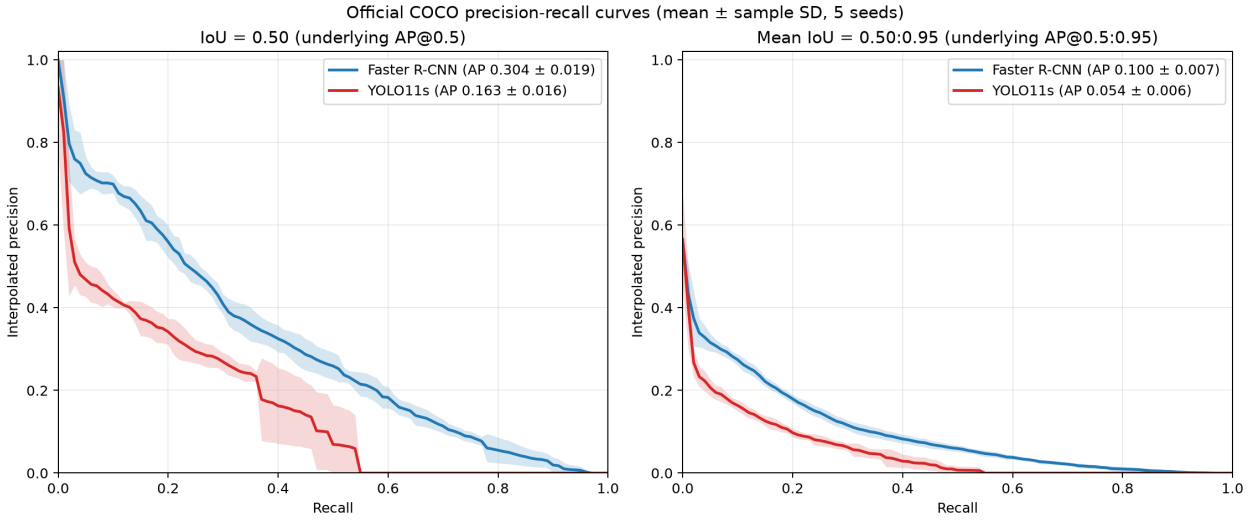


Figure 1a. Official five-run PR curves. Lines and bands show equal-run mean and sample SD, including the low-score YOLO11s run.

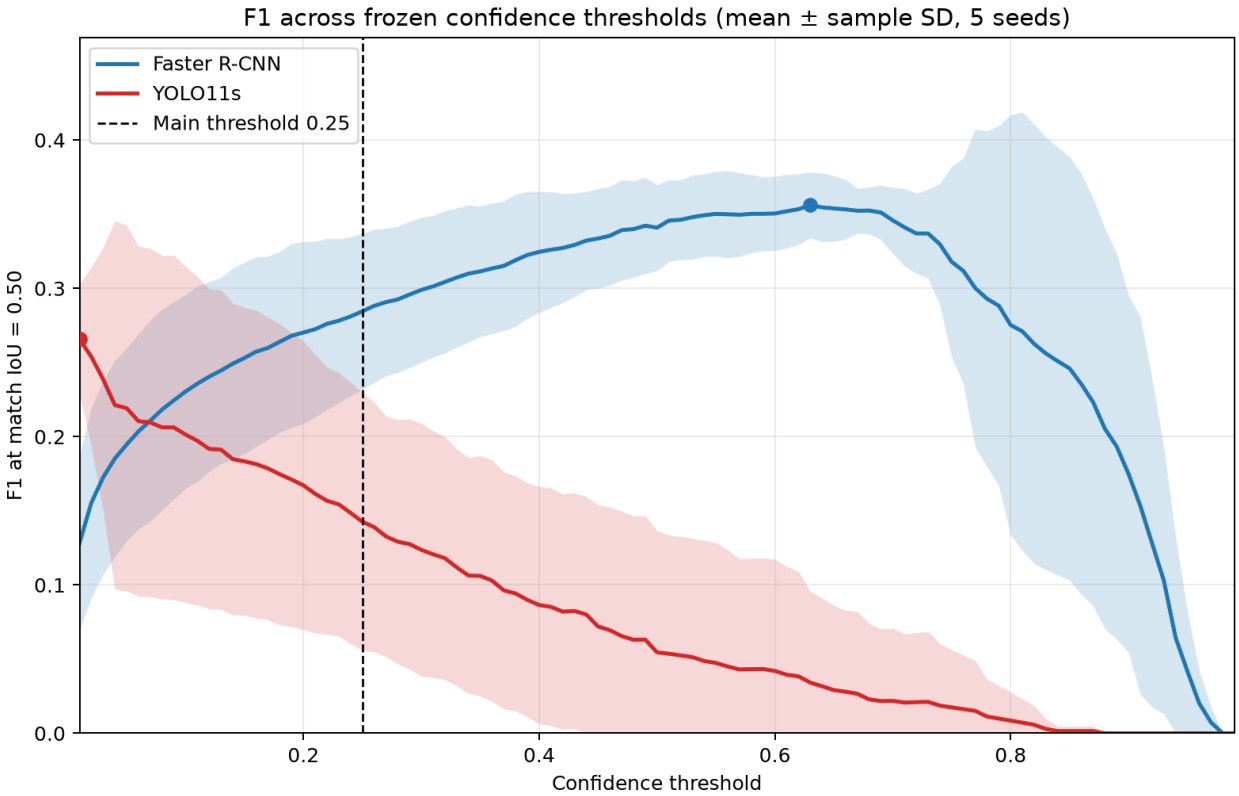


Figure 1b. Exploratory five-run F1 versus raw threshold. Test-set maxima are descriptive and were not used for threshold selection.

Historical validation selection used n=3 and chose 0.69 for Faster R-CNN and 0.05 for YOLO11s. Applying those thresholds unchanged to five test runs gave precision/recall/F1 0.3624 +/- 0.0581, 0.3507 +/- 0.0463, and 0.3511 +/- 0.0184 for Faster R-CNN, versus 0.2524 +/- 0.1418, 0.1948 +/- 0.1110, and 0.2192 +/- 0.1233 for YOLO11s. Seed 271 also emitted no detection at 0.05 and contributed the declared zeros.

The post-hoc validation sensitivity selected 0.70 for Faster R-CNN and 0.01 for YOLO11s from all five validation runs. Applied unchanged to the test set, precision/recall/F1 was 0.3686 +/- 0.0632, 0.3388 +/- 0.0567, and 0.3458 +/- 0.0222 for Faster R-CNN, versus 0.2631 +/- 0.0360, 0.2925 +/- 0.0886, and 0.2657 +/- 0.0367 for YOLO11s. Faster R-CNN's mean precision, recall, and F1 advantages remained but weakened. Mean FP/image was 0.2205 versus 0.3104, reversing the historical-threshold ordering. This is post-hoc validation sensitivity, not a prospectively frozen operating point.

4.3 Internal exact-score false-positive regimes

Five-run FROC sensitivity was higher for Faster R-CNN at each prespecified budget: 0.2776 versus 0.1799 at 0.125 FP/image, 0.3664 versus 0.2664 at 0.25, 0.4858 versus 0.3821 at 0.5, 0.6000 versus 0.5075 at 1, and 0.6978 versus 0.6090 at 2 (Figure 2). These are descriptive equal-run means; aligned inferential intervals are separately available with the transportability analysis.

The 0.00001 candidate floor permits all runs to reach 1 FP/image. YOLO11s seed 137 nevertheless ends at 1.9907 FP/image, so the aggregate at 2 FP/image remains a lower-bound observation. Assigning that run the mathematical maximum sensitivity of 1.0 gives a YOLO11s aggregate upper bound of 0.6963, below Faster R-CNN's observed 0.6978. The internal ordering therefore cannot reverse under this bound for these frozen runs; this is not a population-level uncertainty statement or evidence of a terminal plateau.

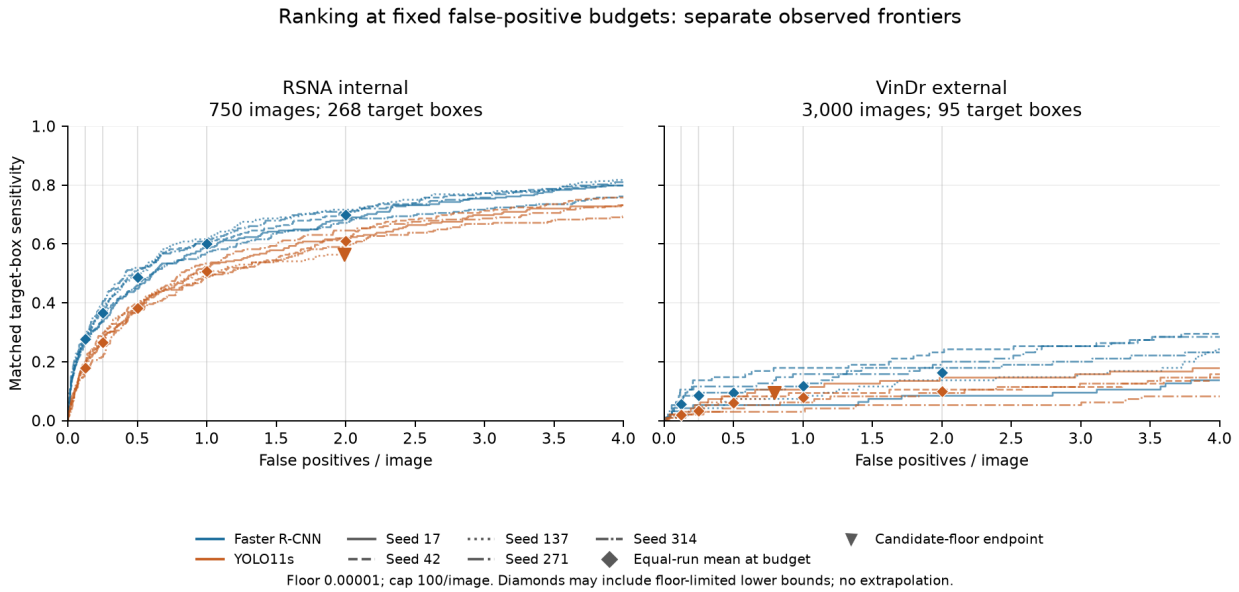


Figure 2. Internal and external observed exact-score FROC, with all five runs and their candidate-floor endpoints. Curves have finite retained support at floor 0.00001 and cap 100. External YOLO11s seed 137 is floor-limited at 1 and 2 FP/image, where the missing-support bound permits reversal. Faster R-CNN cap saturation further limits support; no unrestricted frontier ordering is established.

4.4 Training-procedure uncertainty and conditional tests

The table reports the primary training-procedure differences and separately the secondary checkpoint-conditional p-values. Differences are Faster R-CNN minus YOLO11s; the corresponding absolute endpoint estimates are reported in Section 4.1. Every row contains 750 images from 323 patient clusters. `Runs A/B` gives the eligible Faster R-CNN/YOLO11s trained-run counts.

Table 3. Historical internal uncertainty, with distinct inferential targets.

Endpoint	Runs A/B	Conditioning	Difference (95% training-procedure CI)	Seed 271	Holm p, conditional on observed checkpoints
Precision at 0.25	5/5	Unconditional	-0.1024 (-0.2423, 0.0553)	Both runs; YOLO contributes zero	**0.0020**
Recall at 0.25	5/5	Unconditional	0.4843 (0.3830, 0.5830)	Both runs; YOLO contributes zero	**0.0014**
F1 at 0.25	5/5	Unconditional	0.1419 (0.0559, 0.2290)	Both runs; YOLO contributes zero	**0.0014**
Conditional IoU	5/4	Conditional on a matched detection	-0.0236 (-0.0585, 0.0089)	Faster defined; YOLO undefined	0.1064
Conditional Dice	5/4	Conditional on a matched detection	-0.0171 (-0.0420, 0.0066)	Faster defined; YOLO undefined	0.1064
mAP@0.5	5/5	Unconditional	0.1416 (0.1013, 0.1845)	Both ranked bundles contribute	**0.0208**
mAP@0.5:0.95	5/5	Unconditional	0.0453 (0.0313, 0.0599)	Both ranked bundles contribute	**0.0342**

In Table 3, “Unconditional” means that an endpoint does not condition on a matched detection. All intervals still condition on the fixed training data, recipe, threshold policy and retained candidate support.

Under the primary estimand, the recall, F1, and both AP intervals remained wholly above zero. Fixed-threshold precision did not: its interval crossed zero, so it did not support a training-procedure difference. The small checkpoint-conditional precision p-value instead says that the observed checkpoints favored YOLO11s

at the common numerical cutoff across patient clusters. The CI and p-value differ because the former also resamples trained runs; neither arithmetic result is wrong, and they do not test the same target. Conditional localization excluded missed annotations and remained inconclusive.

4.5 Standardized inference timing

The matched v1 protocol measured Faster R-CNN at 20.91 FPS and YOLO11s at 57.56 FPS. Pooled median image latencies were 47.20 and 17.29 ms, respectively (Figure 3). The table pools three complete 100-image repetitions per frozen checkpoint; dispersion is image-latency IQR, not training-run SD.

Table 4. Measured implementation timing and model size on the reporting laptop.

Matched v1 metric	Faster R-CNN	YOLO11s
Total timed elapsed, 300 calls (s)	14.3466	5.2124
FPS (300 / total elapsed)	20.91	57.56
Median image latency (ms)	47.20	17.29
Q1–Q3 image latency (ms)	46.78–48.00	16.68–17.88
IQR image latency (ms)	1.22	1.19
Total parameters	43,256,153	9,428,179
Training-trainable parameters	43,030,809	9,428,163

All 600 timed-image checks agreed exactly with ordinary inference in counts, category labels, boxes, and scores. These are technical repetitions of one checkpoint per detector, not a new training-replicate analysis. The machine-readable source is `results/tables/inference_timing_v1.csv`; full protocol, individual repetitions, raw intervals, and provenance are indexed in `docs/COMPUTE_TIMING.md`.

The matched throughput ratio was 2.75 in favor of YOLO11s, which had 78% fewer parameters. This compares measured implementations on one machine, not architecture families or clinical workflow speed.

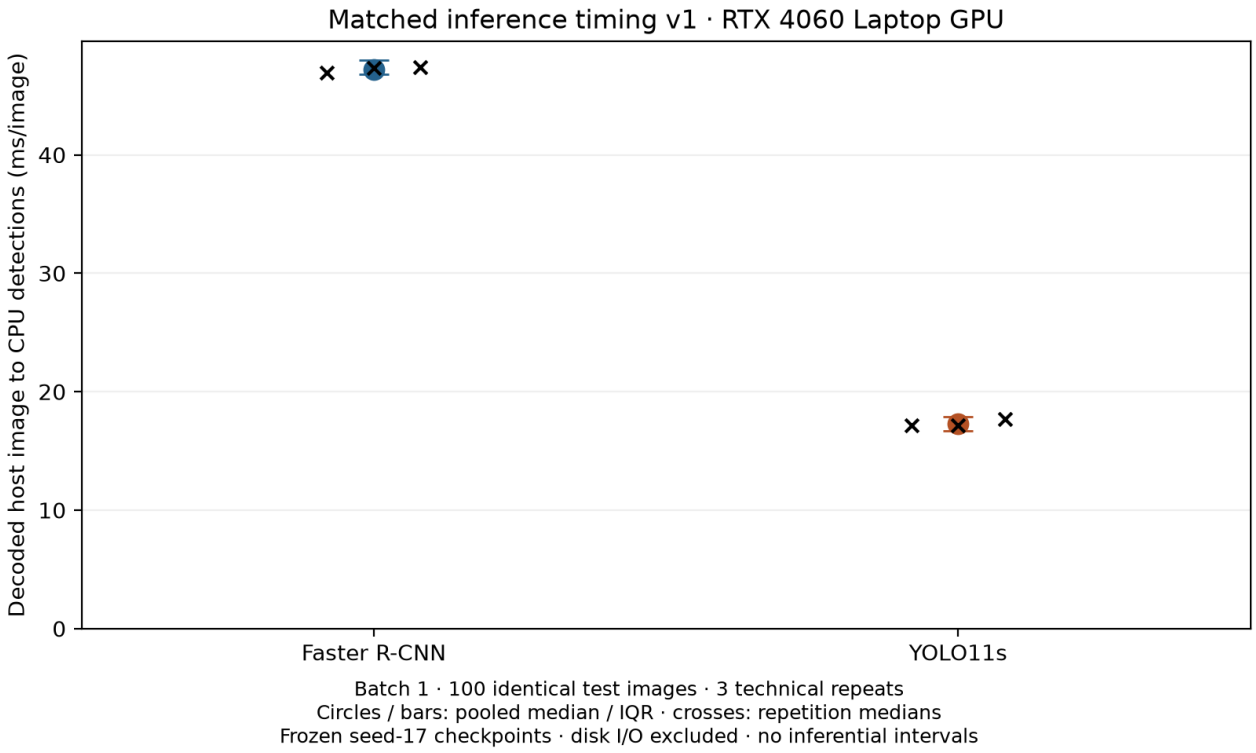


Figure 3. Matched decoded-host inference timing on the reporting laptop. Three repetitions per primary checkpoint are technical measurements; latency dispersion is not training-run uncertainty.

4.6 External cohort and ranking performance

All 3,000 official test images passed the adapter and all ten checkpoint runs completed. The strict target occurred in 84 images (2.8%), with 95 boxes; 2,916 images had no strict-target box. These counts describe target support, not pneumonia prevalence or independent patient counts. They contrast with 169 positive internal-test images and 268 boxes, and limit the precision of external estimates.

Both AP endpoints retained the direction of the equal-run internal ordering, but absolute AP collapsed for both pipelines (Table 5). External AP@0.50 contrasts and AP@0.50:0.95 contrasts had positive marginal training-procedure intervals. This supports a relative difference under the declared estimand, not successful absolute generalization. No particular dataset or numerical implementation factor is identified as the cause.

Table 5. Equal-run AP means +/- sample SD (n=5 per detector); external contrasts are A minus B with marginal 95% training-procedure intervals.

Endpoint	RSNA A	RSNA B	VinDr A	VinDr B	External difference [95% CI]
AP@0.50	0.304224 +/- 0.018896	0.162612 +/- 0.016174	0.002505 +/- 0.001380	0.000501 +/- 0.000272	0.002004 [0.000331, 0.006689]
AP@0.50:0.95	0.099502 +/- 0.006681	0.054168 +/- 0.006030	0.000618 +/- 0.000170	0.000119 +/- 0.000077	0.000499 [0.000075, 0.001719]

4.7 Failure of frozen operating-point transport

Historical RSNA validation thresholds of 0.69/0.05 transported poorly (Table 6). External precision, recall, and F1 were near zero for both pipelines. The lower external FP/image values accompanied sharply reduced emission and recall; they do not establish improvement. All four primary operating-metric contrast intervals included zero: neither a positive mean gap nor preserved AP ordering establishes the same external ordering at the frozen operating points.

Table 6. Historical n=3 validation-selection policy applied unchanged to all five runs in each cohort. Values are equal-run means; full sample SDs and run counts are in Supplementary Section S9. External contrasts have marginal 95% training-procedure intervals. Emission rows are descriptive without intervals.

Historical-policy endpoint	RSNA A	RSNA B	VinDr A	VinDr B	External difference [95% CI]
Precision	0.362419	0.252403	0.010830	0.004271	0.006559 [-0.008360, 0.025664]
Recall	0.350746	0.194776	0.018947	0.004211	0.014737 [-0.004880, 0.045485]
F1	0.351144	0.219206	0.013580	0.004214	0.009366 [-0.006376, 0.031685]
FP/image	0.232000	0.151733	0.047667	0.027067	0.020600 [-0.001735, 0.045672]
Detections/image	0.357333	0.221333	0.048267	0.027200	Descriptive
Images with detections (%)	25.573333	14.213333	4.260000	2.140000	Descriptive

The separately transported post-hoc n=5 RSNA policy used 0.70/0.01. External precision/recall/F1 was 0.011367/0.018947/0.014046 for Faster R-CNN and 0.004499/0.010526/0.006277 for YOLO11s. FP/image was 0.044200 versus 0.060067, reversing the historical-policy ordering within the external cohort, as it did internally. These descriptive sensitivities do not select a replacement policy. Neither policy restores external recall. The proportion of images with detections also changed ordering across datasets under the secondary policy; ordering is endpoint- and policy-specific.

YOLO11s seed 271 emitted no external detection at 0.05 and remained in the five-run zero-valued precision/recall/F1 summaries. At the secondary cutoff of 0.01 it emitted 54 detections, all false positives against the strict target. Its AP still contributed. No inconvenient run was removed or downweighted.

At the common AP support, mean detections/image changed from 16.326133/1.080800 internally to 11.502800/0.217200 externally. The equal-run percentage of images with no such detections rose from 7.466667%/62.373333% to 8.333333%/87.073333%. Per-run score distributions and counts are reported together in the supplement; these emitted-population summaries are not probability calibration and exclude missed targets. Historical FP32 versus external bfloat16 YOLO inference was examined separately in the secondary sensitivity below; score shifts still cannot be assigned to a single dataset or pipeline factor.

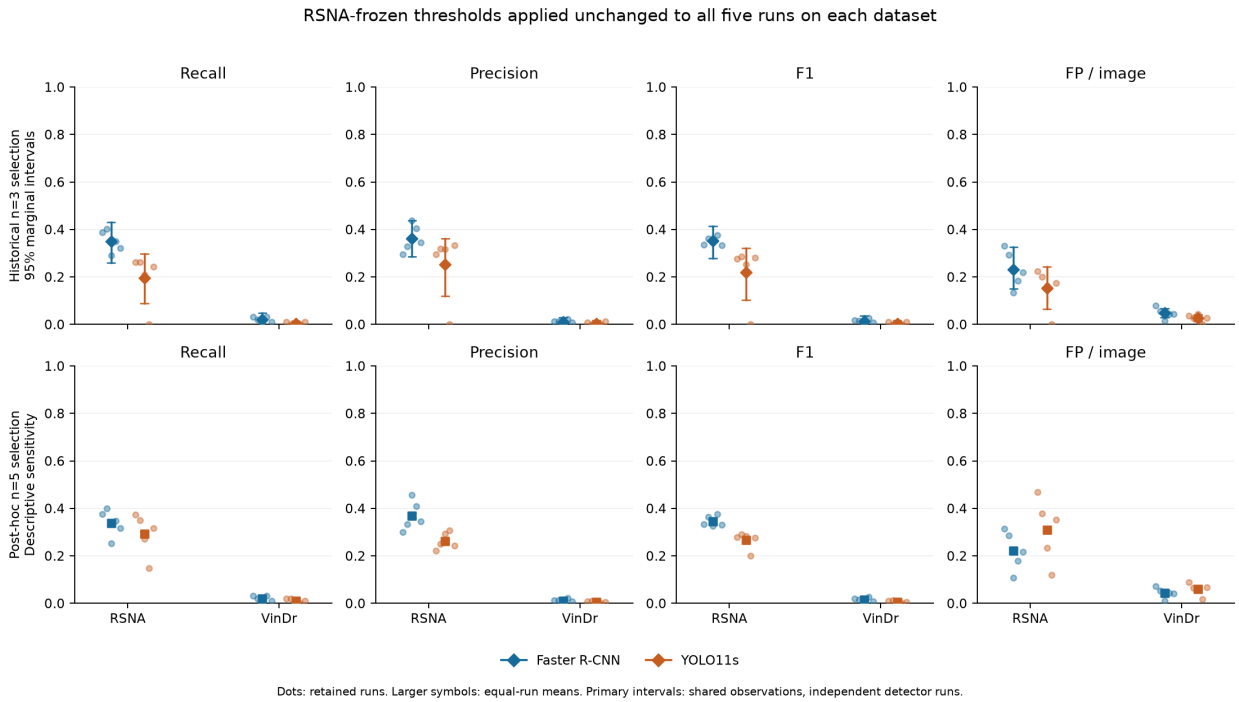


Figure 4. Transport of the frozen historical thresholds, with all retained runs and marginal training-procedure intervals; the separately labeled n=5 validation policy is descriptive post-hoc sensitivity. No VinDr threshold was selected.

4.8 External FROC and uncertainty boundaries

Observed external FROC means retained the internal direction at every specified budget (Figure 2; Table 7), with smaller raw gaps. Four of the five external contrast intervals included zero; only the interval at 0.25 FP/image was wholly positive. These marginal intervals do not establish a simultaneous ordering across budgets.

Table 7. External observed exact-score sensitivities (equal-run means, n=5) and A-minus-B marginal 95% training-procedure intervals. Asterisks identify YOLO11s aggregates containing a floor-limited lower-bound contribution at 1 and 2 FP/image; the missing-support bound permits reversal there. Faster R-CNN cap saturation is an additional support limit. No interpolation or extrapolation was used, and intervals do not recover unobserved candidate support.

FP/image budget	VinDr A sensitivity	VinDr B sensitivity	External difference [95% CI]
0.125	0.056842	0.018947	0.037895 [-0.002128, 0.092453]
0.25	0.086316	0.033684	0.052632 [0.005345, 0.113471]
0.5	0.094737	0.061053	0.033684 [-0.014829, 0.092312]
1	0.117895	0.077895*	0.040000 [-0.020939, 0.108700]
2	0.164211	0.098947*	0.065263 [-0.005663, 0.149396]

YOLO11s seed 137 ended at 0.793333 FP/image and was floor-limited at budgets 1 and 2. Replacing that run's observed sensitivity with the mathematical maximum of 1.0 gives conservative aggregate upper bounds of 0.258947 and 0.280000, respectively, above Faster R-CNN's observed 0.117895 and 0.164211. Thus, unlike the internal bound, the external missing-support bound does not rule out an ordering reversal beyond retained candidates. These are support bounds, not confidence bounds; bootstrapping cannot restore unobserved support. The floor for these original results remained frozen. The separate post-hoc retention diagnostic in Section 4.11 did not replace this analysis.

Faster R-CNN reached the collection cap on 3,000/2,635/3,000/3,000/209 images for seeds 17/42/137/271/314; no YOLO11s image reached it. This substantial cap saturation is another limitation on the observed frontier, not corrected by the YOLO floor bound. The cap for this analysis remained frozen. No unrestricted or full-frontier external FROC superiority is established.

Checkpoint conditioning also changed the comparison. For the seed-17 pair, external AP contrast intervals included zero, and observed FROC sensitivity favored YOLO11s at the three largest budgets. Those secondary results are available in the complete interval table and cannot replace the primary training-procedure interpretation. One checkpoint does not represent an entire training procedure.

4.9 Secondary YOLO numerical-path sensitivity

Under the harmonised internal path, mean AP@0.50 was 0.158412 and mean AP@0.50:0.95 was 0.053137, compared with historical FP32 means of 0.162612 and 0.054168. Mean recall was 0.191791 at the historical cutoff and 0.285821 under the post-hoc policy. Equal-run FROC sensitivity changes ranged from -0.005970 to +0.002239. Observed internal detector orderings at these endpoints were unchanged. Seed 271 retained nonzero AP@0.50 of 0.151761 and a maximum score of 0.040771, remaining below the historical cutoff.

These aggregate performance changes were modest, although post-NMS candidate composition changed and predictions were not numerically equivalent. Detailed score distributions, emissions and box/score agreement are in Supplementary Section S10. Harmonisation left severe external performance and operating-point transport failure intact; it reduces the numerical-path concern without identifying the causes of the remaining cross-dataset differences. Historical primary values and their uncertainty analyses remain unchanged.

4.10 Post-hoc failure decomposition and domain-change sensitivity

The equal-run fractions of strict reference boxes with compatible candidates only below the historical threshold were 0.549474 for Faster R-CNN and 0.176842 for YOLO11s. Fractions without any retained candidate at IoU at least 0.25 were 0.181053 and 0.602105, respectively. These describe saved output; they cannot distinguish absent proposals from floor, NMS or cap removal.

At historical thresholds, mean fractions of strict false positives assigned to overlap with another local finding were 0.536561 and 0.377949, respectively. Those fractions were defined for 5 Faster R-CNN runs and 4 YOLO11s runs; the zero-emission YOLO11s run remained in all count summaries and the full analysis. Overlap describes disagreement with the strict reference definition; it does not measure performance under an expert-adjudicated alternative target or establish that detections are clinically correct or labels should be merged.

After positive-frequency standardization, mean custom weighted AP was 0.108684 for Faster R-CNN and 0.055072 for YOLO11s; mean weighted precision was 0.086639 and 0.062173, respectively. Recall was unchanged by construction. The reweighted values remained well above the external values. This restricted case-mix calculation neither attributes a fraction of failure to prevalence nor proves stable conditional distributions.

Table 8. Post-hoc external-minus-internal changes and marginal 95% intervals. A is Faster R-CNN; B is YOLO11s. The final column is the difference in changes, not a subtraction of independently estimated confidence limits. All endpoints had 2,000 valid draws and no undefined draws. FROC remains restricted to retained support. Larger absolute losses from a higher internal baseline do not establish poorer intrinsic architectural robustness.

Endpoint	A change [95% CI]	B change [95% CI]	A-minus-B change [95% CI]
AP50 change	-0.301719 [-0.366983, -0.232342]	-0.162111 [-0.224218, -0.103034]	-0.139607 [-0.179174, -0.101154]
FROC at 0.25 FP/image change	-0.280102 [-0.367012, -0.188567]	-0.232734 [-0.305797, -0.159547]	-0.047368 [-0.105416, 0.023884]
Historical recall change	-0.331799 [-0.411787, -0.240161]	-0.190566 [-0.292613, -0.081589]	-0.141233 [-0.250947, -0.051971]

Both detectors had negative change intervals at all three endpoints. The difference-in-changes interval included zero for FROC, while AP and historical recall showed larger absolute decreases for Faster R-CNN. Per-run categories, SDs, overlap classes, support diagnostics and prevalence intervals appear in Supplementary Section S11. This analysis reused the original thresholds and retained candidates; the separate retention controls are reported below.

4.11 Controlled retention and display diagnostic

Lowering the collection floor increased YOLO11s' maximum one-to-one coverage at canonical match IoU from 0.181053 to 0.324211 (change 0.143158; marginal interval [0.082605, 0.210538]). For Faster R-CNN, the combined floor/cap change increased coverage by 0.025263 [0.006186, 0.052002]. None of the retention arms changed historical-threshold recall. Header windowing changed retained geometry, but historical-recall change intervals included zero for both detectors. On the full external cohort, lowering only the AP evaluation floor left mean AP50 at 0.002698 for Faster R-CNN and 0.000888 for YOLO11s. All arm results, including unchanged and adverse outcomes, are retained in Supplementary Section S12. These measurements establish sensitivity to the tested controls, without validating a new operating rule or explaining the total cross-dataset loss.

4.12 Limits of threshold repair on retained outputs

Table 9 and Figure 5 show that low external recall was not simply a misplaced numerical cutoff. The best external cutoff at the stated budget remained far below internal performance. Unavailable geometry increased substantially for both pipelines; a lower cutoff cannot create a missing compatible box.

Table 9. Recall accounting at 0.25 FP/image: equal-run means [marginal 95% CI], rounded to three decimals. Components use original retained candidates and the strict reference ontology. The assignment gap was zero in every baseline run; the three displayed components therefore sum to one before rounding. Separate interval endpoints do not sum to one.

Cohort / detector	Cutoff ceiling	FP-budget barrier	Unavailable geometry
RSNA / Faster R-CNN	0.366 [0.299, 0.439]	0.607 [0.530, 0.679]	0.027 [0.012, 0.044]
RSNA / YOLO11s	0.266 [0.206, 0.335]	0.519 [0.380, 0.628]	0.214 [0.124, 0.335]
VinDr / Faster R-CNN	0.086 [0.036, 0.150]	0.482 [0.395, 0.574]	0.432 [0.335, 0.521]
VinDr / YOLO11s	0.034 [0.009, 0.070]	0.147 [0.083, 0.221]	0.819 [0.736, 0.890]

Maximum geometric coverage decreased from 0.973134 internally to 0.568421 externally for Faster R-CNN, and from 0.785821 to 0.181053 for YOLO11s. The corresponding paired changes were -0.404713 [-0.494324, -0.305426] and -0.604768 [-0.693854, -0.502586]. The larger YOLO11s geometric loss does not establish an architecture-level robustness law. The difference in cutoff-ceiling changes remained uncertain: -0.047368 [-0.109369, 0.024273]. The FP-budget barrier was smaller externally in absolute recall units because less compatible geometry remained available; this is not evidence of improved ranking.

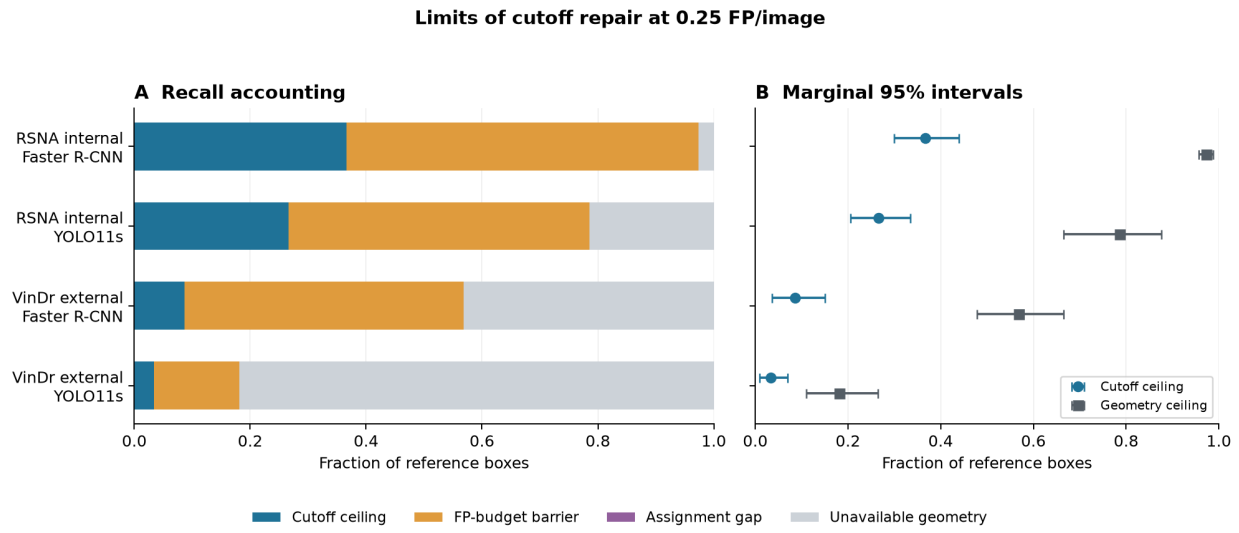


Figure 5. Fixed-output repairability on both complete cohorts. A: cutoff ceiling, FP-budget barrier, assignment gap and unavailable geometry at 0.25 FP/image. B: cutoff and geometric ceilings with joint-bootstrap marginal intervals. The assignment component is zero for these baseline runs. These are conditional annotation-based bounds, not causal shares or an evaluated deployment policy.

All 580 imposed interventions passed their prespecified integrity checks. The 40 score-power controls preserved every exact frontier, although recall at an unchanged numeric cutoff could change. Complete score permutation reduced external cutoff-ceiling means to 0.002105 for Faster R-CNN and 0.013333 for YOLO11s while leaving geometric coverage unchanged. Removing all individually incompatible candidates raised those means to 0.568421 and 0.181053; even this annotation-informed positive control could not exceed retained geometry. Complete compatible-candidate removal eliminated coverage. All intermediate, null and adverse outcomes are available in S13; these software interventions validate diagnostic behavior, not the clinical causes of the real shift.

5. Discussion

The repairability audit provides a concrete reason to avoid treating the external collapse as a score-cutoff problem alone. Within the evaluated candidate sets, strictly monotone score rescaling cannot change the best recall under a fixed FP budget. Reranking or removing false positives can release budget-blocked detections, but recovering lesions outside the geometric ceiling requires changing candidate generation, retention, localisation or the reference task. The latter possibilities are distinct research interventions; this study does not select a successful repair or validate a new ontology.

The internal comparison makes the diagnosis more informative than the external ceiling alone: available geometric coverage contracted sharply, especially for the evaluated YOLO11s pipeline. The output controls demonstrate that the audit distinguishes score ordering from lost geometry. Nevertheless, the accounting identity and monotone invariance are elementary properties, and FROC already supplies the cutoff envelope. The scientific contribution is the reproducible empirical characterization, coupled uncertainty and bounded repair interpretation, rather than a new general-purpose detection metric.

The central finding is the separation between relative ordering and performance transport. Faster R-CNN retained higher observed equal-run AP on VinDr, while both pipelines' AP and the recall of their RSNA-selected operating rules fell severely. **Preserved ordering is not preserved performance** here refers to the observed equal-run AP direction. It does not mean that every external endpoint resolves the same detector difference, that the complete FROC frontier was observed, or that external generalization succeeded.

Internally, a shared numerical cutoff suggested higher YOLO11s precision, whereas ranking curves and detector-specific validation selection gave different comparisons. The post-hoc five-run selection also reversed FP/image ordering relative to the historical policy. Externally, both policies produced very low recall and emission; lower FP/image was therefore an incomplete account of their behavior. Confidence scales are not interchangeable across detectors, and higher AP does not imply better performance at every operating point. Score-scale mismatch is established prior knowledge (Kuzucu et al. 2024); the contribution is its controlled examination alongside training variability, false-positive support, measured efficiency, and external transport.

The retained low-score run illustrates why stochastic training variability matters. Its nonzero internal AP coexisted with zero emissions at conventional cutoffs; excluding it would conceal behavior of the evaluated recipe. Likewise, image/patient-only uncertainty for fixed checkpoints answers a narrower question than resampling both observations and independently trained detector runs. The external seed-17 sensitivity and the historical internal precision analysis show why these interpretations should remain separate.

Measured implementation costs provided another distinct comparison. YOLO11s' smaller parameter count and higher matched-boundary throughput coexisted with lower AP for the studied pipelines. Timing on one machine does not establish an architecture-family efficiency law or clinical workflow benefit. The comparison controls several data and evaluation factors but does not isolate architecture from optimization, normalization, precision, losses, or native API behavior, and it does not compare fully tuned native recipes. Prior RSNA comparisons (Wu et al. 2024; Chinnam et al. 2026) give context without supplying interchangeable scores.

VinDr external testing was deliberately frozen without adaptation, so it measures transport of established pipelines and rules. Its strict local opacity ontology differs from RSNA's reference standard. Acquisition, hospital/site, country, population, annotation ontology and preprocessing effects remain confounded. Internal numerical-path harmonisation produced modest aggregate changes and preserved the major observed comparisons. This tested numerical-path change did not reproduce the magnitude of the observed cross-dataset loss. It does not establish that every remaining difference is a dataset effect or attribute the collapse to any one factor. Nor would combining labels after seeing results repair that inference. The adverse result motivates separately designed future transport studies; it does not justify post-hoc tuning on this external test set.

The post-hoc decomposition makes these limits more specific. Compatible below-threshold candidates coexist with localization failures and missing retained support, in different proportions across pipelines. Overlap with

other local findings makes annotation mismatch plausible, while preserving the strict evaluation. Population weighting lowers internal precision and custom AP without reproducing the external results. Together these observations constrain explanations; they do not partition causation among prevalence, ontology, image appearance or model behavior. The paired cross-domain changes quantify absolute loss, conditional on retained candidates, rather than rank intrinsic detector robustness. The FROC difference-in-changes interval remains compatible with either direction.

The subsequent controlled diagnostic confirms that some apparent candidate absence depends on the collection floor. Recovering low-score geometric coverage did not recover historical-threshold recall, and relaxing the AP evaluation floor left absolute performance very low. The tested header-window mapping likewise did not establish improved threshold recall. These bounded interventions narrow specific implementation explanations; they do not isolate population, acquisition or annotation effects, or demonstrate that a different preprocessing policy would generalize.

6. Limitations

Data and reference standards. The hardware-scoped 5,000-study RSNA subset is not a deployment-prevalence sample, and only two disclosed detector pipelines were evaluated. Coarse expert boxes and the nonspecific opacity target do not establish pneumonia diagnosis. Technical annotation checks are not an independent radiologist rereading. RSNA source accrual dates and broader clinical/acquisition variables are unavailable locally; header age requires the nominal-year assumption. Min-max conversion and resizing do not reproduce vendor display processing. VinDr has a non-identical ontology, only 84 strict-target-positive images and 95 boxes, and no defensible patient grouping. Image-level external intervals may be too narrow if repeat patients exist. The numerical-path sensitivity narrows one implementation concern; dataset, site, population, annotation and preprocessing differences remain inseparable in this design.

Training and selection. Five runs per detector are a coarse empirical sample of training variability, conditional on one fixed training dataset. Disabled augmentation may disadvantage YOLO11s relative to its usual recipe. Optimization, normalization, and precision asymmetries limit architecture-only interpretation. Historical thresholds used three validation runs; five-run selection remains post-hoc sensitivity and does not rewrite that provenance. Equal-weight F1 does not encode measured clinical harms, and the YOLO five-run choice reaches the selection grid's lower boundary. Internal hypotheses and analyses were retrospective; the external freeze was a local prespecification.

Evaluation and uncertainty. AP and FROC depend on retained support and postprocessing. The internal floor-bound no-reversal result does not transport to VinDr's upper budgets. External YOLO candidate-floor limits and substantial Faster R-CNN detection-cap saturation preclude a full-frontier conclusion. Marginal intervals offer no familywise coverage, omit threshold-selection uncertainty, and cannot recover missing support. The later cross-domain change intervals are exploratory marginal contrasts, with no multiplicity-adjusted interaction test. Conditional IoU/Dice exclude misses and use unequal numbers of defined runs. Historical internal YOLO accuracy inference used FP32 while external inference used verified bfloat16 AMP. The secondary internal sensitivity found modest aggregate changes but non-identical candidate composition; it is not an equivalence test, and historical TF32 flags were not recorded. It does not identify a dataset-only cause for the remaining shifts. No calibration map was fitted. Detection-level D-ECE remains descriptive, emitted-population dependent, binning/support dependent, and not patient-level clinical-risk calibration.

Post-hoc diagnostic scope. Error categories and best-IoU summaries depend on post-NMS retained output and cannot diagnose pre-NMS proposal failure. Other-finding overlap lacks expert adjudication and does not validate a merged target. Population weighting holds internal conditional case mix and observed external positive frequency fixed; its custom AP is not an official COCO metric. These analyses were specified after the original external results, and neither threshold-selection uncertainty nor missing candidate support is recovered by the joint bootstrap. The controlled diagnostic used a fixed negative-control sample; its FP/image cannot characterize the complete external population. Geometric coverage uses annotation-informed assignment and is not an achievable operating point. The specific header window is not validated as clinical vendor display processing, and zero-containing change intervals do not establish equivalence. The separate full-cohort broader-support inference proposal remains unexecuted.

Repairability scope. The new ceilings are conditional on original candidate membership and source reference boxes. They do not characterize unseen proposals or calibration methods that change ranking or ties. The selected FP/image budget is an analytical setting, not a clinically validated tolerance. Five observed runs cannot represent all training variability, and the joint marginal intervals provide neither multiplicity-adjusted findings nor missing-support recovery. The controlled output changes were specified after the original outcomes

and are software interventions, not simulations of real clinical domain-shift mechanisms. No TIDE/LRP performance comparison or independent radiologist adjudication was performed. A third independently evaluated cohort would be needed to assess how broadly these repairability patterns generalize.

Secondary analyses and compute. Corruption, acquisition stress, and Grad-CAM analyses use one primary checkpoint per detector and limited samples. Synthetic stored-array changes do not validate scanner/dose models; Grad-CAM and its controls do not establish causal or clinically appropriate reasoning. The runtime comparison uses one checkpoint per detector, different AMP dtypes, 100 images, and one hardware/software state. Technical repetitions do not estimate retraining or hardware uncertainty, and disk I/O and workflow costs are excluded. Incomplete profiler GFLOPs remain supplementary.

Clinical and reporting scope. No prospective evaluation, reader study, validated clinical deployment, diagnostic utility, patient benefit, or complete fairness analysis was established. Demographic representation is not subgroup performance evidence. Portable CI verifies public evidence without licensed images; full scientific replay requires authorized data and matching checkpoints. Restricted VinDr image-linked derivatives remain private, and the ten checkpoint binaries have no public release URL. Retraining is not promised to be bitwise identical, including the recorded CUDA ROI Align backward limitation. Institutional ethics/consent applicability, the final submission release identifier, source-reporting gaps, and a participant-flow diagram remain unresolved in the reporting crosswalk.

7. Conclusion

A leakage-aware, multi-run comparison separated ranking performance, raw-score behavior, validation-selected operating points, false-positive regimes, training variability, and measured implementation efficiency. Frozen VinDr testing preserved the observed equal-run AP ordering while exposing severe loss of absolute performance and failure of internal operating rules to transport well. Uncertain endpoint contrasts and finite FROC support limit broader ordering claims. Post-hoc error, annotation-overlap and weighting analyses sharpen the description of failure without isolating its causes. The repairability audit further shows that global cutoff adjustment alone has limited headroom under a fixed FP budget and cannot overcome absent retained geometry. Internal/external contrasts and controlled output changes support this restricted computational conclusion without identifying a clinical cause. Reporting these evaluation objects separately makes the comparison informative without implying clinical validation or universal superiority of either detector family.

8. Declarations

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Competing interests. The authors declare no financial or nonfinancial competing interests related to this work.

Ethics and data use. This study was a secondary analysis of previously collected, de-identified RSNA/NIH and VinDr-CXR datasets. The authors did not recruit participants. No separate institutional ethics approval, exemption or written determination was sought or obtained from Imam Khomeini International University or K. N. Toosi University of Technology for this analysis. The official VinDr-CXR documentation reports approval of the original study by the institutional review boards of Hanoi Medical University Hospital and Hospital 108 (Nguyen et al. 2021). These source-study approvals are distinct from a determination for the present secondary analysis.

Consent. The authors did not obtain new participant consent. The VinDr-CXR dataset documentation reports a consent waiver for the original retrospective study because clinical care and workflow were unaffected and identifying information had been removed (Nguyen et al. 2021). No separate consent determination was obtained for the present analysis.

Ethics review status. The authors have not yet confirmed whether their institutions or the target journal require a separate ethics review or consent determination for this secondary use of RSNA/NIH and VinDr-CXR data.

Author contributions (CRediT). Pouyan Delivandani: Methodology, Software, Data curation, Investigation, Formal analysis, Visualization, Writing - original draft, Writing - review & editing. Mohammad Amin Hajialirezaei: Formal analysis, Visualization, Writing - review & editing.

Data, code, and model availability. Code, configurations, reproduction instructions and public aggregate results are available in the project repository. Repository-authored software is licensed under AGPL-3.0-only; source datasets and third-party components retain their respective terms. RSNA/NIH source data are accessed through the RSNA challenge under its provider terms. VinDr-CXR is accessed through its credentialed PhysioNet release and data-use agreement. Raw source data and restricted VinDr image-linked derivatives are not redistributed by this repository. The authors have chosen to keep all ten trained model checkpoints unpublished at initial submission; no public checkpoint download is provided. Reproduction using the exact trained checkpoints requires access to those unpublished files. The public repository supports portable code and aggregate-artifact verification without them.

Submission version. A public commit or archive identifier for the version submitted to the journal has not yet been selected.

Patient and public involvement. Patients and members of the public were not involved in developing the research question, designing or conducting the study, interpreting the results, or planning dissemination. This study used previously collected datasets.

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